

Star Topology Network CYB-2.1.1.1

1. Which device is the central device?
S1
2. How many end devices are connected to the central device?
4
3. How many physical connections are shown?
4
4. What makes this design a star topology?
All devices connect to a “star”, a concentration or central node.
5. What would happen to the connected devices if the central switch failed?
They would all be disconnected.
6. What would happen if only one PC cable were disconnected?
The others would still be connected but that PC would be cut-off.

Mesh Topology Network CYB-2.1.1.2

1. How many switches are in the mesh design?
3
2. How many links connect the switches to one another?
3
3. How many possible physical paths appear to connect S1 and S2?
2
4. Which alternate path could connect S1 and S2 if their direct link were unavailable?
S1-S3-S2
5. How does the mesh design provide more physical redundancy than a simple star?
Distributed responsibility, not one single point of failure.
6. What would happen to the overall design if one inter-switch cable were disconnected?
Not much, it'd be rerouted through what's still connected.
7. Why does a mesh require more cables and ports than a star?
More interconnecting nodes need more cables, Star only has one connecting node.